kW to Amps Chart

Current for a given real power at the common three-phase and single-phase voltages.

Three-phase amps are kW times 1,000, divided by 1.732 times the voltage times the power factor. At 480 V and a power factor of 0.9, each kW draws about 1.34 A, so 100 kW is 134 A. The three-phase table below is worked at 0.9; the note under it says how to adjust for any other value.

Three-phase, power factor 0.9

kW	208 V	240 V	480 V	600 V
1	3.08	2.67	1.34	1.07
2	6.17	5.35	2.67	2.14
5	15.4	13.4	6.68	5.35
7.5	23.1	20.0	10.0	8.02
10	30.8	26.7	13.4	10.7
15	46.3	40.1	20.0	16.0
20	61.7	53.5	26.7	21.4
25	77.1	66.8	33.4	26.7
30	92.5	80.2	40.1	32.1
40	123	107	53.5	42.8
50	154	134	66.8	53.5
75	231	200	100	80.2
100	308	267	134	107
150	463	401	200	160
200	617	535	267	214
250	771	668	334	267
500	1,542	1,336	668	535
750	2,313	2,005	1,002	802
1,000	3,084	2,673	1,336	1,069

Amps per phase. For another power factor, multiply by 0.9 and divide by yours: at 0.8, the figures are 12.5 percent higher.

Single-phase, power factor 1.0

kW	120 V	240 V
0.5	4.17	2.08
1	8.33	4.17
1.5	12.5	6.25
2	16.7	8.33
3	25.0	12.5
4	33.3	16.7
5	41.7	20.8
7.5	62.5	31.3

kW	120 V	240 V
10	83.3	41.7
15	125	62.5
20	167	83.3

Amps for a resistive load such as a heater. For a load with a lower power factor, divide by it.

How the values are calculated

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I = kW × 1000 / (1.732 × V × PF)  [three-phase]
I = kW × 1000 / (V × PF)  [single-phase]
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Every figure in the chart comes from the formulas above. Nothing is copied from a published table, so any cell can be checked by hand.

What this chart does not tell you

This is steady-state current at the stated power factor. It is not the current to size a conductor or a breaker from: a continuous load is taken at 125 percent, motors are sized from the code tables and not from nameplate power, and starting current is several times the running figure.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/kw-to-amps-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.